Research Questions and Answering Plan

Working document - not supervisor-approved. Research-question wording provisional pending D-RQ-01 and D-RQ-02.

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0. How to read this document

This document does one job: it states each research question in its current form, shows why that form exists, argues that the question is genuinely open, and sets out in detail how it will be answered. It is the connective tissue between the Gap and Research Questions chapter (Chapter 3, drafted) and the Methodology and Plan chapters (Chapters 4 and 6, deliberately not started).

Four conventions are used throughout, and they are load-bearing.

Evidence identifiers. References of the form `E4`, `E12`, `A08-04` point to rows in the canonical record of the 2026-08-05 supervision call. That record is a **machine-derived transcript with inferred, undiarized speakers**, generated by automatic speech recognition from Hebrew audio and not yet reviewed by the participants. Every attribution below is therefore phrased as *attributed to X in the machine-derived record*. No statement here should be read as a supervisor having approved a specific final wording. Action `A08-01` — verifying the post-edit wording against Ali's own saved working draft rather than against the transcript — remains outstanding, and until it closes, every question in this document is a best reconstruction.

Decision identifiers. References of the form `D-RQ-01` point to rows in the Research-Question Decision Pack. All ten decisions in that pack are recorded as **Pending**. Section 6 lists them, together with six further decisions this document proposes; the proposed ones are marked as such and are not logged IDs.

Artifact deferral. On 2026-08-05 the instruction recorded as `A08-04` was to *think about, but not start*, the definition of the research artifact per question. This document complies. Wherever an artifact is discussed it is presented as a set of **options with their trade-offs**, and no option is selected, ranked, or recommended. Selecting one is supervisor territory and is deliberately left open.

Evidence boundary. No claim of accuracy, macro-F1, effort reduction, generalization, superiority, or clinical performance appears anywhere in this document, because none is currently supportable. The boundary is restated in full in Section 7 and is not weakened by anything in between.

Planning blocks used throughout

Requirement `E14`, attributed to Iris and confirmed by Arnon in the machine-derived record, is that the plan be written in approximately three-month, semester-aligned blocks over a **three-year** horizon, never month by month. This document uses the following block scheme.

Block	Approximate span	Character
B0	Aug – Sep 2026	Pre-candidacy proposal block
Y1-A	Oct 2026 – Jan 2027	Year 1, autumn semester
Y1-B	Feb – Jun 2027	Year 1, spring semester
Y1-S	Jul – Sep 2027	Year 1, summer block
Y2-A	Oct 2027 – Jan 2028	Year 2, autumn semester
Y2-B	Feb – Jun 2028	Year 2, spring semester

Y2-S	Jul – Sep 2028	Year 2, summer block
Y3-A	Oct 2028 – Jan 2029	Year 3, autumn semester
Y3-B	Feb – Jun 2029	Year 3, spring semester
Y3-S	Jul – Sep 2029	Year 3, summer block

These boundaries follow a conventional academic-year structure. They have **not** been verified against either university's actual calendar, candidacy rules, or submission deadlines. That verification is decision `D-RQ-10`, which currently has no named owner. Any September or October date mentioned below is an internal working target, not a confirmed university deadline.

1. U-RQ — The main research question

1.1 The question

> ***U-RQ.*** *How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human–AI co-reasoning?*

Wording status. Provisional. The wording above is the live text refined during the 2026-08-05 working call, reconstructed from a machine transcript, and it has not been checked against Ali's own saved working draft (`A08-01`, open). The governing decision is **D-RQ-01** — *use this wording as the working baseline* — recorded as **Pending**; **D-RQ-02** — *keep the exactly-three sub-question structure beneath it* — is also **Pending**. Five supervisor requirements still bear on this sentence: **E4** (the question must not be a description of the proposed solution), **E5** (the word "reused" must not appear in the headline), **E6** (the question must be narrowed explicitly to variability identification and classification), **E7** (retain "reliable"; lean toward dropping "auditable", "transferable" and "end-to-end"), and **E8** (the "expert" versus "human" terminology choice, which the record leaves open with a leaning toward "expert").

Two specific wording problems are unresolved and should be settled at sign-off rather than inherited silently:

- The headline says *human judgment*; SQ2 and SQ3 say *expert judgment*. The four questions currently disagree with one another. `E8` leans toward "expert" throughout, which would require amending U-RQ and SQ1. This is proposed below as `D-RQ-11`.
- `E6` asked for narrowing *explicitly to variability identification and classification*. The live wording says *variability exploration*, which names the broader activity of which identification and classification are parts. Whether "exploration" discharges `E6` is a judgment the supervisors should make, not one this document should assume. Proposed below as `D-RQ-12`.

1.2 What the supervisors required about this question

The main question was rewritten live, word by word, for roughly half an hour of the 2026-08-05 call. The following requirements produced its present shape.

Requirement	What it demanded	Effect on the current wording
E4 (attributed to Arnon)	The draft conflated the proposed agentic-AI solution with the actual research question; cut verbal padding; distinguish the core question — the ability to identify variability — from technical and implementation matters.	The question now names an **activity** (variability exploration in guideline operationalization) rather than a system. Agentic AI appears as the setting whose reliability is at stake, not as the contribution.
E5 (attributed to Iris)	Remove "reused" from the main question; reuse concerns how knowledge is generalized and folded back in, which is a sub-question matter.	"Reused" is gone from the headline. It survives in SQ2 (governed reuse) and SQ3 (transfer).
E6 (attributed to Iris, echoing Arnon)	The question was too broad — "domain-specific artifacts" and "human–AI	"Variability exploration in guideline operationalization scenarios" replaced the

	reasoning" in general; narrow it explicitly to variability identification and classification.	general artifact framing. Whether this is narrow enough remains open (see D-RQ-12).
E7 (attributed to Iris, Arnon and Ali jointly)	Keep "reliable"; lean toward dropping "auditable", "transferable" and "end-to-end" from the headline.	Only "reliable" survives as the quality target. Auditability became a property of the SQ2 artifact; transferability became SQ3's subject matter.
E13 (attributed to Iris)	The research questions themselves are not to be written as domain-specific; domain framing belongs in the methodology chapter.	The question names no domain. Software/modelling and healthcare appear as instantiation contexts elsewhere, not in the question.

The requirement set is coherent: E4 removes the solution, E5 and E7 remove sub-question concerns and adjectives, E6 narrows the activity, E13 removes the domain. What remains is intended to be a question about an activity and a quality target, and nothing else.

1.3 Why it is open

The main question is open for two reasons that are distinct and must not be collapsed.

First, its three constituent unknowns are each unresolved, and Sections 2.3, 3.3 and 4.3 argue that independently. If it is not settled when an agentic assessment process should request expert judgment, nor how such judgment must be represented to remain contestable and scoped, nor what makes a judgment transferable beyond the context that produced it, then it is not settled how human judgment can be captured, governed and used to support this activity at all.

Second, and separately, the composition is itself untested. Answering three questions in isolation does not entail that their answers compose into a working lifecycle. An intervention policy tuned to surface important uncertainty produces a particular *distribution* of captured judgments — skewed toward hard, contested, atypical cases. A representation designed for those judgments inherits that skew. A transfer classification learned from a skewed store may generalize differently from one learned from a representative sample. Whether the three stages reinforce or undermine one another is an empirical property of the composed lifecycle, not a corollary of the three separate results. No existing work, to our knowledge, has composed these three stages and measured the interaction — a statement that the frozen literature searches must confirm or refute before it can be asserted as a finding.

There is a third source of openness which is easy to overlook: **"reliable" is not an operationalized construct for this task.** Reliability of a classifier is well defined. Reliability of a *co-reasoning arrangement* — in which a system proposes an interpretation of a deviation, an expert accepts, corrects or rejects both the verdict and the reasoning behind it, and the resulting judgment may later inform other cases — has no accepted measurement definition. Agreement with a reference standard is insufficient, because in this task the reference standard is itself expert judgment and legitimate expert disagreement is not error but the very phenomenon under study. Defining what reliability means here is part of answering the question, not a preliminary to it.

Finally, a boundary that keeps this honest. The question does **not** presuppose that captured human judgment improves variability exploration. Whether it does, under which controls, and at what cost in expert effort, is an empirical outcome. A well-conducted result showing no improvement, or improvement only under conditions too narrow to be useful, answers the question.

1.4 What answering it requires

The study. U-RQ has no study of its own. It is answered by the conjunction of Study 1, Study 2 and Study 3 plus a **cross-study synthesis** that is not reducible to any of them: the synthesis must state what the three results jointly establish about the composed lifecycle, where they conflict, and what boundary conditions they place on one another.

The method. Design Science, applied per research question, following the structure confirmed in `E13` (attributed to Iris, referencing Prof. Penina's course): for each question a study, an artifact, a design rationale, and an evaluation. At the U-RQ level the method is the synthesis of three design cycles plus an explicit argument about their composition. The unit of analysis at this level is the lifecycle as a whole, not any single component.

The data and evidence needed. The union of the three studies' evidence, plus one thing none of them produces alone: at least one end-to-end trace in which a specific captured judgment can be followed from the intervention decision that elicited it, through its representation and governance, to a reuse or non-reuse decision in a later case, with the effect of that decision observable. Without such traces the composition claim is unevaluable.

The artifact — options only, none selected (A08-04). Three framings are visible, and they differ in what would count as the doctoral contribution:

1. **No separate U-RQ artifact.** U-RQ is answered by the composition of the three sub-question artifacts plus the synthesis argument. *Strength:* avoids inventing a fourth deliverable to satisfy symmetry. *Weakness:* leaves the main question without anything of its own to inspect, which may read as thin.
2. **An integrating reference model.** A single artifact binding intervention, capture and reuse into one specified lifecycle with defined interfaces between the stages. *Strength:* makes the composition claim concrete and inspectable. *Weakness:* risks becoming an architecture diagram, which is exactly the solution-versus-question conflation `E4` warned against.
3. **A design theory.** A set of abstracted design principles and boundary conditions, derived from the three studies, stating what any system attempting this lifecycle must provide and under what conditions it will fail. *Strength:* most clearly a knowledge contribution rather than an engineering one. *Weakness:* hardest to evaluate; generalizations from three studies in at most two domains may be over-reach.

These are not ranked. Whether U-RQ needs an artifact at all is proposed below as `D-RQ-16`.

The evaluation design. At U-RQ level, evaluation is the defensibility of the synthesis: whether each sub-question's answer is supported by its own evidence at its own stated strength; whether the composition claim is supported by end-to-end traces rather than inferred from the parts; and whether the boundary conditions stated are the ones the evidence actually supports.

Success criteria.

- Each of SQ1, SQ2 and SQ3 has an answer, at a stated and honest evidence strength, including "readiness-only" where that is what the evidence supports.
- The composed lifecycle has been exercised end to end on real captured judgments, and the interaction between the stages is characterized rather than assumed.
- "Reliable human–AI co-reasoning" has been given an operational definition, and that definition has been applied to real data at least once.

Failure criteria — what would defeat the question. A distinction matters here. A *negative empirical result* — governed judgment does not improve variability exploration under the conditions tested — answers the question and is reportable. What would *defeat* the question is different, and there are three such outcomes:

- **The premise fails.** If agentic variability exploration turns out to agree with adjudicated expert judgment closely enough that selective human involvement has no measurable role on the constructs studied, then human judgment is not needed to support the activity and the question, as posed, is idle.
- **The construct fails.** If "reliable human–AI co-reasoning" resists operational definition — if no measure survives scrutiny as anything other than agreement with one expert panel — the question cannot be answered as asked and must be reformulated.
- **The composition fails structurally.** If the three stages prove to have incompatible requirements — for example, if the judgment distribution that a defensible intervention policy produces is systematically

unusable for transfer analysis — then the lifecycle does not compose, and the honest result is a decomposition claim, not an integration claim.

All three are reportable results. None is a project failure. All three should be stated as possible outcomes in the proposal rather than discovered late.

1.5 The plan to answer it

Phase	Produces	Evidence yielded	Preconditions and gates	Block
U-0 Wording closure	Final agreed text of U-RQ and SQ1–SQ3; logged `D-RQ-01`/`D-RQ-02`; resolution of the terminology and narrowing questions	A stable question set that downstream chapters can be written against	`A08-01` verification against Ali's saved working draft; supervisor sign-off at the 2026-08-12 meeting or later	B0
U-1 Construct definition	A written operational definition of "reliable human–AI co-reasoning" for this task, with candidate measures and their known weaknesses	The measurement basis every later study reports against	U-0 complete; literature searches `QL-01`–`QL-05` executed	B0 → Y1-A
U-2 Composition specification	A statement of the interfaces between the three stages: what SQ1 hands SQ2, what SQ2 hands SQ3, what SQ3 returns	Makes the composition claim falsifiable in advance rather than reconstructed afterwards	Study 1 and Study 2 designs stable enough to have interfaces	Y1-B
U-3 End-to-end trace capture	Instrumented traces following individual judgments from elicitation to reuse decision	The only evidence that speaks to composition rather than to parts	Real captured judgments exist; EXP-005 labelling under way	Y2-A → Y2-B
U-4 Cross-study synthesis	The synthesis chapter: joint findings, conflicts, boundary conditions, and the answer to U-RQ at its supportable strength	The answer to the main question	Studies 1–3 reported	Y3-B
U-5 Thesis assembly	Introduction (written last, per E13) and final integration	—	U-4 complete	Y3-S

1.6 Risks and mitigations

Risk	Consequence	Mitigation
The wording is never formally logged and drifts between documents	Chapters 2 and 4 are written against a moving target; late rework	Treat `D-RQ-01`/`D-RQ-02` as blocking for Chapters 2 and 4; keep every version in the change log rather than overwriting
The reconstructed wording differs from what was actually agreed on 2026-08-05	The proposal argues for a question the supervisors did not endorse	`A08-01`: verify against Ali's saved working draft, not against the transcript. Until then, every use of the wording carries the provisional flag
"Reliable" is never operationalized and remains an adjective	The main question cannot be answered, only gestured at	Phase U-1 is an explicit early deliverable with its own output, not an assumption
Composition is asserted from three separate results	The main question's distinctive contribution collapses into three papers with no thesis	Phase U-2 specifies the interfaces in advance; phase U-3 requires end-to-end traces as a separate evidence type
The medical instantiation never opens	Perceived narrowing of the doctorate	The question is domain-neutral by

		construction (E13) and is fully answerable in software/modelling alone. Healthcare tests its outer boundary; it is not a premise. See the Plan A / Plan B contract in Section 4.6

1.7 Current status

What exists. A complete working draft of Chapter 3 argues the gap and states all four questions with an openness argument for each. The Three-Study Contract maps each sub-question to one study with method, units of analysis, measures, exclusions, dependencies, fallback and exit criterion. The Research-Question Decision Pack sets out the wording options and the ten decisions. VEGO-AI itself — the four-agent framework published by the supervisors at MODELS '26 — provides a concrete, instrumented setting in which the gap is demonstrably real.

What is honestly missing. No `D-RQ` decision is logged; all ten are Pending. `A08-01` is not done, so the exact agreed wording is not confirmed. "Reliable" has no operational definition. The literature searches `QL-01`–`QL-05` are protocol-ready and **not executed**, so no novelty or review-completeness statement can be made and every literature-gap statement in Chapter 3 is a candidate claim rather than a finding. Chapters 1, 2, 4 and 6 are not started — for Chapters 2 and 4 this is compliance with `E15` and `A08-04`, not slippage.

2. SQ1 — Selective intervention

2.1 The question

> ***SQ1.*** *When and how, in variability exploration scenarios, should an agentic assessment system request human judgment so that important uncertainties are addressed without unnecessary expert burden?*

Wording status. Provisional, under **D-RQ-02** (the exactly-three structure), recorded as **Pending**. The clause *"in variability exploration scenarios"* was inserted on instruction `E7`, which also directed that SQ1 otherwise stay close to its pre-call form; the rest of the sentence is unchanged from the baseline in the Decision Pack. Two requirements still bear on it: **E8** (SQ1 says *human judgment*, while SQ2 and SQ3 say *expert judgment*; the record leans toward "expert" throughout, which would require amending this sentence) and **E4** (the question must remain about *whether and when to ask*, not about the mechanism that does the asking).

2.2 What the supervisors required about this question

Requirement	What it demanded	Effect
E7 (attributed to Iris, Arnon and Ali jointly, during the live edit)	Insert "variability exploration scenarios" into SQ1; otherwise keep SQ1 close to the original.	Binds SQ1 to the same activity as the main question, so the sub-question cannot drift into general human-in-the-loop design.
E4 (attributed to Arnon)	Separate the proposed agentic solution from the research question.	SQ1 asks *when and how* an assessment process should request judgment. It does not ask how to build a review queue. Its outputs are criteria and policies with measured consequences, not components.
E8 (attributed to Iris and Arnon; recorded as an open choice)	Prefer "expert" over "human"; in the medical instantiation the source is a physician, in software/modelling it is course- or team-level engineering rigour.	Not yet applied to SQ1. Pending as `D-RQ-11`.

The question also carries its own evaluation criterion inside its wording, and this was deliberate. Success is two-sided — important uncertainties addressed **and** expert burden bounded. A policy that achieves coverage by flooding the expert fails SQ1 by definition. This is what makes it falsifiable rather than aspirational.

2.3 Why it is open

Both extreme policies are known failures, and that is the least interesting part of the argument. Review everything and the automation is pointless; expert attention is spent uniformly, which means it is spent mostly on cases that did not need it. Review nothing and the interpretive step — deciding whether a deviation is a legitimate alternative, an error, a domain convention, a defect in the guideline, or a genuine ambiguity — is forfeited, and that step is the heart of the task.

What is genuinely unresolved is the space between them, and it is unresolved for four reasons.

Importance is not confidence. The signals an agentic process emits about its own uncertainty are not evidence about whether a case matters. A system can be confidently wrong about a deviation whose misclassification would propagate into a guideline revision, and unconfident about a deviation that is inconsequential either way. Threshold-only triggers inherit the calibration weaknesses of the underlying model and answer the wrong question. What "important" means for a *deviation from an operationalized guideline* is not defined by any construct we are aware of — a claim the frozen searches `QL-01` and `QL-04` must test.

The decision is sequential and its cost is irreversible. The system must decide, during its own operation and before knowing the outcome, that a particular uncertainty warrants interrupting a person. Attention spent cannot be recovered and cannot be spent again. This is a budgeted sequential decision problem, whereas the human-oversight literature characteristically assumes an idealized reviewer of unbounded availability. Under a real budget, the question is not "should this case be reviewed?" but "is this case worth more of the remaining budget than the cases still to come?" — and the cases still to come are not yet observed.

Agentic processes multiply the decision points. A multi-step, tool-using process does not produce one judgment per artifact; it produces many candidate judgments at several stages, some of which are inputs to others. Intervening early is cheaper and less informed; intervening late is better informed but may be too late to change anything downstream. Where in a multi-step process the intervention belongs is a question the literature on single-decision oversight does not reach.

There is a bootstrapping problem, and it does not dissolve. Determining which uncertainties are important is itself an expert judgment. Any evaluation of an intervention policy therefore requires a reference standard of importance that only experts can supply — which is precisely the resource the policy is trying to conserve. This is not an implementation inconvenience; it constrains what evidence about SQ1 can exist and in what order it can be obtained, and it is why SQ1's plan below is gated on the same independent-label programme as SQ3.

Note what this section does **not** do, per `E4`: it does not argue that a particular triage design would resolve any of the four points. Whether it would is a research outcome.

2.4 What answering it requires

The study. Study 1 — Selective-intervention architecture, per the Three-Study Contract.

The method. Design-science construction with requirements-to-artifact evaluation; intervention-policy modelling; architecture traceability; bounded scenario analysis; analytical burden modelling; and structured literature synthesis once `QL-01`–`QL-05` have been run. The later phases add retrospective policy replay over a frozen corpus and, once reviewers exist, a prospective reviewer study.

Units of analysis. Assessment events; uncertainty and importance signals; review candidates; intervention decisions; dosage modes; routing outcomes; timeouts; reviewer tasks.

Data and evidence needed. Existing VEGO-AI and human-layer requirements, event catalogues, execution traces, specifications, tests and controlled fixtures; the literature corpus; supervisor decisions on eligibility, dosage, timeout and routing. Synthetic fixtures remain mechanism tests only and cannot support an effect claim. For the evaluative phases: adjudicated importance labels, and reviewer interaction data with real timings.

The artifact — options only, none selected (A08-04). Three candidates are visible from what already exists:

4. **An intervention architecture** — listener, triage and routing components generalized from the existing queue-building pipeline. *Strength:* lowest friction; matches the Study 1 language already in the Decision Pack. *Weakness:* reads as an engineering deliverable and would need an explicit generalizable claim layered on it to count as a doctoral contribution rather than a component diagram — precisely the E4 risk.
5. **A trigger taxonomy paired with a dosage policy** — a domain-neutral classification of *why* a case is escalated, plus a threshold or budget rule. *Strength:* taxonomies are straightforwardly evaluable for precision and recall. *Weakness:* a taxonomy alone is thin and probably needs a decision procedure attached to stand as a full artifact.
6. **A parameterized decision framework** — thresholds and escalation rules as parameters rather than one fixed policy instance. *Strength:* gives SQ1 and SQ3 methodological symmetry. *Weakness:* risks blurring into SQ3's evaluation artifact unless "decide when to intervene" is kept analytically separate from "measure the effect".

Not ranked; not a shortlist.

The evaluation design. Two-sided by construction. Axis one: coverage of important uncertainty — of the cases an adjudicated panel considers important, what fraction did the policy route for review? Axis two: expert load — how many cases were routed in total, and what reviewer time did they consume? A policy is characterized by a point on this plane; a family of policies traces a curve. The evaluation compares policy variants on that curve, first by retrospective replay against the frozen corpus and adjudicated labels, then prospectively with real reviewers. Pre-registration of the comparison and of the importance standard precedes any run.

Success criteria.

- Every SQ1 requirement maps to an inspectable artifact and a validation.
- Burden and uncertainty controls are explicit, and their behaviour under timeout and escalation is specified and tested.
- The coverage–load plane is measurable, and the policy family occupies a **non-degenerate** region of it — that is, there exists at least one policy that materially reduces load while retaining coverage that blanket review would achieve.
- Unsupported effect claims remain excluded; the supervisors accept or correct the stated boundary.

Failure criteria — what would falsify or defeat the question.

- **Degenerate trade-off.** If every reduction in load costs a proportional loss of coverage — if the curve is a straight line from blanket review to no review — then selective intervention buys nothing over random sampling, and SQ1's premise that *selective* intervention is possible is falsified. This is a genuine, reportable negative result.
- **Unmeasurable importance.** If an adjudicated panel cannot agree on which uncertainties are important — if inter-rater agreement on the importance standard is too low to support a reference set — then SQ1 as posed is not answerable and must be reformulated around a different criterion, such as downstream consequence rather than judged importance.
- **Wrong locus.** If policy variation turns out to matter far less than *when in the multi-step process* the request is made, then the question was posed at the wrong granularity and should be reformulated around placement rather than triggering.

Fallback if dosage effects cannot yet be measured (per the study contract): report the inspectable architecture, the analytical burden model, the scenario validation, the unresolved decisions and the pre-registered empirical test — and keep effectiveness claims excluded.

2.5 The plan to answer it

Phase	Produces	Evidence yielded	Preconditions and gates	Block
S1-1 Requirements	A consolidated, traceable requirement	Requirements-to-artifact coverage	Wording closure (U-0);	Y1-A

consolidation	set for selective intervention, derived from the existing event catalogue and the literature synthesis	baseline	QL-01–QL-05 executed	
S1-2 Importance construct	A written definition of what makes an uncertainty important in this task, with candidate operationalizations	The reference-standard design that every later comparison depends on	S1-1; agreement with supervisors that importance is a construct to be defined rather than assumed	Y1-A
S1-3 Artifact construction	The selected artifact (option chosen by supervisors), specified and instrumented	Inspectable design; trigger and eligibility coverage	Artifact decision taken (A08-04 released); S1-2	Y1-B
S1-4 Analytical burden model	A model projecting review rate and reviewer load under stated policies and corpus characteristics	Review-rate and burden **projections** — explicitly projections, not measured effects	S1-3	Y1-B
S1-5 Scenario and traceability validation	Bounded scenario suite covering eligibility, priority, dosage, routing, escalation and timeout; component-to-requirement traceability table	Mechanism and readiness evidence; escalation and timeout behaviour; decision traceability	S1-3	Y1-B → Y1-S
S1-6 Retrospective policy replay	Policy variants replayed over the frozen corpus against adjudicated importance labels	The first two-sided coverage–load evidence	**Gate: adjudicated importance labels exist.** Depends on the same independent-label programme as SQ3 (see 4.5). Blocked while EXP-005 is at 0 of 24	Y2-A
S1-7 Prospective reviewer study	Reviewer sessions under stated policies, with real timings and real routing decisions	Measured reviewer time, cases per hour, escalation rates	S1-6; reviewers recruited; supervisor-approved reviewer plan	Y2-B
S1-8 Study 1 report	Study 1 write-up; Paper 1 draft on selective intervention in agentic assessment	The answer to SQ1 at its supportable strength	S1-5 minimum; S1-6/S1-7 if the gates opened	Y1-S (readiness report) → Y2-B (full report)

Note the deliberate split at S1-8: a readiness-level Study 1 report can be produced in **Y1-S** from mechanism evidence alone, and upgraded in **Y2-B** if the label gate opens. This is the fallback made schedulable rather than hypothetical.

2.6 Risks and mitigations

Risk	Consequence	Mitigation
The importance reference standard never materializes because it depends on the same scarce expert attention SQ1 studies	S1-6 and S1-7 never start; SQ1 stalls at readiness evidence	Accept the readiness-level answer as a legitimate Study 1 outcome (the contract's stated fallback); pre-register the empirical test so it is executable the moment labels exist; keep the importance standard small — a subset sufficient for a reference set, not full corpus coverage
The artifact reads as engineering rather than research	E4's conflation criticism recurs at examination	Keep the openness argument (2.3) and the artifact strictly separate in the writing; require any artifact option to carry an explicit generalizable claim
Reviewer recruitment fails or reviewers	No prospective evidence; leakage risk if	Reviewer plan approved by supervisors in

are not independent	reviewers have seen system output	advance; blind sheets that hide system classifications; adjudication by a third party
Policy space explored too narrowly	A degenerate trade-off is concluded when the space simply was not searched	Pre-register the policy family and its parameter ranges before replay; report the searched region explicitly
Corpus too small for policy discrimination	Differences between policies are within noise	Report this as a power limitation rather than as a null effect; the current corpus is 27 variability patterns across 4 settings, which is small

Plan A / Plan B. SQ1 is not medically gated. Under both plans it is executed in the software/modelling setting. Under Plan A only, medical workflow requirements may be used as a **design stress test** — and only after expert review, never as data. Nothing in SQ1 depends on the medical track opening.

2.7 Current status

What exists today. A rule-based selective-intervention mechanism is implemented and merged: `selective_intervention_policy.py` together with `human_review_queue.py`, applying a conjunctive filter over the fourth agent's own uncertainty signals — `requires_human_review`, `Undetermined`, low or medium confidence, and `flag_for_guidelines_update`. On the current offline corpus this routes **11 of 27** variability patterns to a review queue. The corpus snapshot reports 4 settings and 179 cases locally; the published foundation paper reports 178 case models, and that discrepancy is recorded, not reconciled here. Schemas, tests and a component-to-evidence traceability table exist.

What is honestly missing. There is no definition of importance, so the routing rule cannot yet be evaluated for whether it routes the *right* 11. There is no burden model, no dosage or budget policy, and no escalation or timeout semantics beyond what the current filter implies. There are no reviewers, no reviewer timings, and no adjudicated importance labels — EXP-005 stands at **0 of 24** generalization-safe labels against a threshold of at least 20. The literature synthesis that would establish which of the four openness arguments in 2.3 is actually unaddressed by prior work has not been run. No artifact has been selected. Consequently **no claim of reduced expert burden, improved coverage, or policy superiority is made or supportable.**

3. SQ2 — Governed knowledge reuse

3.1 The question

> ***SQ2.*** *How should expert judgment — including the system's core reasoning — be represented, validated, reconciled, and stored so it can be reused transparently without unsafe generalization or loss of human authority?*

Wording status. Provisional, under **D-RQ-02** (**Pending**). Three requirements bear on it directly. **E9** required tying SQ2 explicitly to *core reasoning* and adding an evaluation-criteria clause in the style of SQ1 and SQ3 — the core-reasoning clause is now present; whether the trailing clause *"without unsafe generalization or loss of human authority"* discharges `E9`'s request for a **correctness-versus-completeness balance** is open. **E8** required splitting SQ2 conceptually into (a) what the expert says and how it is captured and (b) how it transfers to future cases; the current sentence spans both halves but does not visibly separate them, and whether the split should surface in the wording is unresolved. **E10** recorded a mild, non-critical discomfort with the word *"transparently"* — recorded against SQ3, though the word now appears in SQ2; Chapter 3 already flags this relocation as a derived interpretation requiring confirmation rather than inheritance.

Three proposed decisions follow from this: `D-RQ-13` (retain or drop "transparently", and against which question the reservation attaches), `D-RQ-14` (make the `E8` split visible in the wording, or keep one sentence), and `D-RQ-11` (terminology — note that SQ2 already uses "expert", so under `D-RQ-11` it is U-RQ and SQ1 that would move, not this sentence).

3.2 What the supervisors required about this question

Requirement	What it demanded	Effect
E8 (attributed to Iris and Arnon; open choice)	Split SQ2 into (a) what the human says and how it is captured, and (b) how that transfers to future cases. Prefer "expert" over "human" — physicians in the medical domain, course- or team-level rigour in software engineering.	SQ2 now deliberately spans both halves and hands the second over to SQ3 at the point where "may it be reused" becomes "does reuse hold up across contexts". "Expert" is adopted here.
E9 (attributed to Iris)	Tie SQ2 explicitly to *core reasoning*, which appears in the main question but was addressed by no sub-question; add an evaluation-criteria clause matching SQ1 and SQ3 (balance correctness against completeness; avoid unsafe generalization and loss of human authority).	"Including the system's core reasoning" is now inside the question. The trailing constraint clause is present; the correctness/completeness balance is arguably still implicit.
E10 (attributed to Iris; open choice)	Mild discomfort with "transparently".	Word retained pending decision; its placement in SQ2 rather than SQ3 is flagged as derived.
E5 (attributed to Iris)	Reuse belongs in a sub-question, not the headline.	SQ2 is where governed reuse lives.

The reason E9 matters more than it looks: the judgment worth capturing is not a bare label. An expert responds to the argument the agentic process gives for flagging a deviation, and may endorse, correct or reject *that argument*, not only the verdict. A representation that drops the reasoning cannot distinguish "the expert disagreed with the conclusion" from "the expert disagreed with the reasoning" — and those have different implications for whether the judgment should be reused.

3.3 Why it is open

The two mature paradigms for capturing human input both destroy properties that expert assessment judgment requires.

Static knowledge bases capture expert rules explicitly but detach them from the cases that motivated them. What remains is a rule without its grounding, and in this task the grounding is what determines the rule's reach: a ruling that a particular difference between a student's model and a reference solution is a legitimate alternative may hold for that guideline, that course, that modelling language — or far more broadly. Stripped of the case, the rule cannot be scoped.

Weight-absorbing approaches — fine-tuning, preference learning — take the opposite route and absorb human signals into model parameters, where they become uninspectable, unattributable and irrevocable. An institution cannot answer whose judgment is being applied, and an expert cannot withdraw a judgment they no longer endorse.

Expert assessment judgment has four properties that neither paradigm preserves, and each generates an open problem.

It is case-grounded. An expert rules on *this* deviation, in *this* artifact, under *this* guideline. Representations that generalize at capture time discard the evidence that would let a later reader test whether the generalization was warranted.

Its scope is unknown at the moment of capture. This is the crux, and it is genuinely hard. When an expert rules on a case, neither the expert nor the system knows how far the ruling extends — whether it holds for this course, this institution, this modelling language, or the whole class of guideline-operationalization tasks. So any adequate representation must carry an **under-determined** scope that later evidence can narrow or widen, and must record what evidence would do either. We are not aware of an established representation for judgment that treats scope as an evidential variable rather than a fixed attribute — a claim the frozen searches must confirm.

It is contestable, and disagreement is not noise. Two experts may legitimately disagree, and in a thesis whose subject is *variability* this is not an inconvenience to be averaged away — legitimate alternative interpretations are the phenomenon under study. Any reconciliation rule that resolves conflict by majority or by recency is therefore question-begging: it destroys precisely the signal the research is about. What a defensible reconciliation regime looks like when disagreement may be substantive rather than erroneous is open.

It is authority-bearing, and authority is temporal. Institutions need to know whose judgment is applied and whether it is still endorsed. Endorsement can be withdrawn; an expert may revise their view; an institution may change guidelines. A store that cannot represent revocation and propagate it to everything downstream cannot preserve human authority in any meaningful sense. Most knowledge stores model insertion and retrieval, not withdrawal and its consequences.

There is a further point that follows from E9 and is open in its own right. If what is captured includes the system's reasoning and the expert's response *to that reasoning*, then the representation must distinguish at least four states — endorse verdict and reasoning; endorse verdict, reject reasoning; reject verdict, accept reasoning as relevant; reject both — and these have different reuse consequences. Whether that distinction can be elicited reliably from real experts, and whether it changes any downstream decision, is unknown.

Finally, an argument about why this is a research question and not an engineering one: an ungoverned judgment store cannot support claims about *which* judgment produced *which* downstream effect. Without attributable provenance, reuse is not scientifically evaluable at all. The representation is thus a precondition for evidence, not a convenience.

3.4 What answering it requires

The study. Study 2 — Governed judgment lifecycle.

The method. Design-science artifact evaluation; schema and contract validation; source-grounded verification scenarios; conflict and adjudication analysis; authority analysis; provenance inspection; bounded-convergence analysis.

Units of analysis. Judgment records; evidence sources; validations; conflicts; reconciliations; authority transitions; provenance chains; retrievals; reuse proposals; expiries and revocations; convergence outcomes.

Data and evidence needed. Existing judgment and feedback schemas; provenance and memory artifacts; verification and conflict fixtures; tests; governance requirements; and — for anything beyond mechanism validation — **real approved expert judgments**, which do not yet exist.

The artifact — options only, none selected (A08-04).

7. **A governance framework** spanning capture, validation, promotion, storage, retrieval and advice, with schema, signature and conflict rules as its controls. *Strength:* near-verbatim match to the Study 2 description already agreed in the Decision Pack. *Weakness:* "framework" is broad; reviewers may demand a sharper boundary between framework, protocol and process.
8. **A knowledge-representation or schema architecture** — the schema pair plus their provenance links treated as the artifact, i.e. a formal representation model for reusable expert judgment. *Strength:* concrete and checkable now, via schema conformance and provenance completeness. *Weakness:* undersells the retrieval, matching and conflict logic, which is arguably the more distinctive part.
9. **A safe-reuse classification scheme** — classifying a judgment along scope, conflict and match strength to decide whether and how it may be reused. *Strength:* tracks the most distinctive existing mechanism, deterministic and explainable retrieval without embeddings. *Weakness:* narrower; does not obviously cover the validation and governance side without being paired with another option.

Not ranked; not a shortlist.

The evaluation design. Two tiers, and the boundary between them is the human-evidence gate.

Tier 1 — mechanism validation, available now. Schema and contract validity; provenance completeness; validation and reconciliation coverage; conflict and authority-rule coverage; retrieval-explanation completeness; **unsafe-reuse rejection** measured against a constructed adversarial suite of cases that *should* be refused; scope, expiry and revocation enforcement, including whether revocation propagates to everything derived from a withdrawn judgment; bounded convergence.

Tier 2 — human-evidence-gated. Whether experts can supply scope at capture time with acceptable agreement; whether the verdict-versus-reasoning distinction is elicitable and consequential; whether reconciliation rules produce outcomes experts endorse on review.

Success criteria.

- Every SQ2 lifecycle stage, authority rule and unsafe-reuse control maps to an inspectable artifact and a validation.
- The adversarial suite is refused at the specified rate, and every refusal carries an explanation a reader can check.
- Revocation propagates completely and demonstrably.
- The representation demonstrably distinguishes verdict-disagreement from reasoning-disagreement on at least one set of real captured judgments.
- Transparent reuse remains advisory and gated until evidence permits more.

Failure criteria — what would falsify or defeat the question.

- **Scope is not elicitable.** If experts cannot supply the scope of their own rulings with acceptable agreement, the scoped-judgment construct fails and SQ2 must be reformulated around *inferred* scope — a different and harder question.
- **The core-reasoning clause is idle.** If capturing the system's reasoning, and the expert's response to it, changes no reuse decision in any observed case, then `E9`'s clause is empirically inert and should be dropped from the question. Stating this in advance is what makes the clause falsifiable rather than decorative.
- **Governance is self-defeating.** If the controls that prevent unsafe generalization also prevent every reuse that would have been safe — if the scheme refuses everything — then governed reuse as posed offers nothing over no reuse, and the question's premise fails.
- **Conflict is undecidable in principle.** If substantive expert disagreement cannot be distinguished from erroneous disagreement using any evidence available at capture time, then reconciliation cannot be principled, and the honest answer is that judgments must be stored unreconciled with their disagreement preserved — itself a result, but a different one from the question as asked.

Fallback if real judgment records remain unavailable (per the study contract): report mechanism and contract validation and the human-evidence gate; do **not** infer safe reuse from fixtures.

3.5 The plan to answer it

Phase	Produces	Evidence yielded	Preconditions and gates	Block
S2-1 Representation requirements	Requirements for a judgment representation derived from the four properties in 3.3, plus the `E9` reasoning-response states	The specification any artifact option must satisfy	Wording closure; `QL-02` synthesis on knowledge capture and expertise reuse	Y1-B
S2-2 Artifact construction	The selected artifact (supervisor choice), specified with its governance controls	Inspectable design; schema and contract validity	Artifact decision taken; S2-1	Y1-S

S2-3 Adversarial unsafe-reuse suite	A constructed suite of cases that must be refused, with the reason each must be refused	Unsafe-reuse rejection evidence; the strongest Tier-1 result available without human data	S2-2	Y1-S
S2-4 Provenance and authority validation	Provenance-completeness audit; revocation-propagation test; authority-transition coverage	Governance evidence; revocation and expiry enforcement	S2-2	Y2-A
S2-5 Bounded convergence analysis	Analysis of whether repeated capture and reconciliation converges or oscillates under stated rules	Convergence evidence	S2-2, S2-4	Y2-A
S2-6 Scope elicitation study	Real experts supplying scope tags and reasoning responses on real cases; agreement analysis	**Tier 2**: whether the central constructs are elicitable at all	**Gate: real expert judgments exist.** Depends on the reviewer programme shared with SQ1 and SQ3	Y2-B
S2-7 Study 2 report	Study 2 write-up; Paper 2 draft on the governed judgment lifecycle	The answer to SQ2 at its supportable strength	S2-3 to S2-5 minimum; S2-6 if the gate opened	Y2-S

3.6 Risks and mitigations

Risk	Consequence	Mitigation
No real expert judgments ever arrive	SQ2 is answered only at mechanism level	Accept mechanism-and-contract validation as the stated fallback; report the human-evidence gate explicitly rather than inferring safe reuse from fixtures
Fixtures are mistaken for evidence of safe reuse	An unsupportable governance claim enters the thesis	The contract already excludes this by name: claims that governed reuse improves outcomes or generalizes safely **may not** rest on schemas, memory, tests or synthetic fixtures alone
The adversarial suite is written by the same person who designed the controls	The suite tests what the controls already handle	Have the suite reviewed independently, ideally by whoever did not design the artifact; derive cases from the four failure properties in 3.3 rather than from the implementation
Scope elicitation burdens experts so heavily that they disengage	The Tier-2 evidence is never collected	Keep the elicitation instrument minimal; treat elicitation cost as a measured outcome of S2-6, not an overhead to be hidden
SQ2 and SQ3 blur at the scope boundary	Two studies claim the same contribution	Fix the handover explicitly: SQ2 owns *may this be reused*, SQ3 owns *does reuse hold up*. Restated in Section 5

Plan A / Plan B. Under both plans, Study 2 is executed first in software/modelling with independent reviewers and leakage controls. Under Plan A only, a medical judgment lifecycle may follow — and only after an approved clinician and data protocol exist. No part of SQ2 depends on the medical track.

3.7 Current status

What exists today. A schema-validated feedback path with signature-verified attachment; a provenance-tracked, scope-tagged judgment memory (`human_judgment.schema.json`) with explicit promotion rules and conflict handling; and a deterministic, embedding-free, explainable retrieval layer producing graded advice strength. The retrieval layer is advisory only: across the current offline corpus it emitted 8 memory-advice items and changed **zero** classifications, which is the intended, approved behaviour rather than a shortfall. Three reusable memory entries and four resolved feedback items exist. A `reuse_scope` field already distinguishes same-pattern, same-setting, cross-setting and cross-domain reuse.

What is honestly missing. There are **no independent expert judgments** in the store — the three memory entries are same-pattern and are explicitly barred from serving as generalization evidence. The representation has never been exercised against a real expert response to the system's reasoning, so the `E9` distinction is untested. There is no adversarial unsafe-reuse suite and therefore no rejection evidence. Revocation propagation is unvalidated. Convergence is unanalysed. Whether experts can supply scope at capture time is entirely unknown. No artifact has been selected. Consequently **no claim that governed reuse is safe, transparent, or beneficial is made or supportable.**

4. SQ3 — Evaluation and transfer

4.1 The question

> ***SQ3.*** *How can expert judgment be reused and transferred across different guideline-operationalization contexts without unsafe generalization or loss of human authority, first in software/modeling and, when governance and access permit, in healthcare?*

Wording status. Provisional, under **D-RQ-02** (**Pending**). **E10** records that this wording "looks good" as drafted, with a mild, non-critical reservation about *"transparently"* — a word that no longer appears in this sentence but does appear in SQ2 (see 3.1 and `D-RQ-13`). **E12** is the substantive outstanding item: it is a confirmed directive that SQ3 focus on **classifying domain-specific versus broadly transferable** elements, and the current wording does not name that classification. Chapter 3 carries it as SQ3's analytic core in prose. Whether the question's own text should name it is proposed below as `D-RQ-15`. **E5** placed reuse here rather than in the headline, which is why this sentence carries both "reused" and "transferred".

4.2 What the supervisors required about this question

Requirement	What it demanded	Effect
E12 (attributed to Arnon)	Uncertainty in transferring user-involvement patterns is often but not always domain-driven. Failure to define actors and use cases is a **general capability gap**, not a domain-specific one. SQ3 should focus on classifying domain-specific versus broadly reusable or transferable elements.	Reframes SQ3 from "does it transfer?" to "which parts transfer, and how would we tell?". This is the single most consequential correction to SQ3 and is not yet reflected in its wording.
E11 (attributed to Iris; recorded as discussion, not directive)	Start deliberately from the software-engineering domain, not medicine, because domain-transfer behaviour is unproven. A chronic-pain management guideline (Clalit) and an age-related macular degeneration guideline (Soroka) were offered as maximally different examples motivating caution.	Justifies the "first in software/modeling" ordering inside the question and the conditional healthcare clause.
`E10` (attributed to Iris; open choice)	Wording approved as drafted, with a mild reservation about one word.	Wording largely stable; one lexical decision outstanding.
E13 (attributed to Iris)	Preliminary results are software/modelling only; medical is infrastructure-building only.	Bounds what SQ3 may report at proposal time.

`E12` deserves restating because it is the analytic heart of the question. Two judgments can look identical on the surface and have opposite transfer status. A judgment about whether a modeller failed to identify the actors of a described system concerns a capability that is equally hard, and equally improvable, in every domain — so it is a transfer candidate. A judgment about whether a chronic-pain treatment sequence may legitimately deviate from a clinical guideline is bound to clinical context — so it must not travel. The difference lies not in the judgment's form but in **why the uncertainty arose**.

4.3 Why it is open

Transfer claims are currently untestable, not merely untested. Without a criterion separating domain-specific from general-capability uncertainty, "this judgment transfers" has no truth conditions beyond "it happened to work in the second setting I tried" — which does not distinguish transfer from coincidence. Establishing the criterion is prior to measuring anything.

Transfer status is not observable from the judgment. It is a property of the uncertainty's origin, and origin is not recorded by any capture scheme we are aware of. Two judgments with identical text, scope tag and confidence can differ in transfer status. This means the classification cannot be read off the store; it must be derived from evidence about *why* the system was uncertain, which is a different and unsolved elicitation problem.

Honest measurement requires evidence that does not exist. Testing transfer requires independent labels in both the producing and the evaluating context, and leakage controls between them — because a judgment that informed a classification cannot then be used to score that classification. The evidence discipline is demanding, and the current state is zero independent labels.

Negative transfer is possible and unbounded. Reused judgment could actively degrade assessment in a new context — by importing a convention that is wrong there, or by suppressing a deviation that matters there. Nothing in the literature we are aware of bounds negative transfer for this task, and an evaluation that reports only successful transfers is not an evaluation.

Domain distance is unquantified. E11's example — a chronic-pain guideline and a macular-degeneration guideline sharing almost nothing at the domain level — illustrates that "different context" is not a single step size. Two software/modelling settings, two clinical specialties, and one of each are three different distances, and there is no scale on which to place them. Without one, a transfer result cannot be interpolated to any other pair of contexts, which limits what any single transfer study can claim.

Note what is *not* argued here, per E4: no claim that a particular classification scheme would resolve these points. Whether any scheme has predictive validity is the empirical content of SQ3.

4.4 What answering it requires

The study. Study 3 — Evaluation and cross-domain transfer.

The method. A controlled comparative empirical study in software/modelling, followed by a comparative transfer case. Blinded independent labelling; adjudication; paired comparison; workload, error and validity analysis; and replication under Plan A or Plan B.

Units of analysis. Assessment cases; reviewer judgments; changed-versus-unchanged decisions; settings and domains; datasets; artifacts and processes; reviewer panels; required adaptations; governance profiles; cross-context outcomes.

Data and evidence needed. The frozen VEGO-AI software/modelling baseline; the existing 27-pattern evidence; the **24 generalization-safe EXP-005 candidate rows**; independent reviewer returns; adjudicated labels; and then either Plan A approved healthcare evidence or Plan B an authorized second software/modelling context. **EXP-005 currently stands at 0 of 24.**

The artifact — options only, none selected (A08-04).

10. **An evaluation protocol artifact** — a reusable protocol combining paired comparison, leakage tagging and gold-label gating. *Strength:* matches the Study 3 description already agreed, and the results chapter needs it

regardless. *Weakness:* a protocol reads as methodology rather than artifact, which may reintroduce exactly the conflation E4 flagged.

11. **A transfer classification scheme** — extending the existing scope field into a full scheme classifying which elements are domain-specific and which broadly transferable. *Strength:* directly answers E12, so it is grounded in a supervisor instruction rather than invention. *Weakness:* sits logically close to SQ2's scope field; the SQ2/SQ3 boundary would need working out rather than assuming.
12. **A comparative measurement framework** — a domain-neutral instrument for measuring quality, consistency, traceability and effort deltas. *Strength:* the most literal extension of what already runs. *Weakness:* closest to "the existing implementation, renamed"; needs tight scoping to measurement logic alone to read as a distinct contribution.

Not ranked; not a shortlist.

The evaluation design. Pre-registered before any run. Two blind independent reviewers label from sheets that hide system classifications; a third party adjudicates disagreements into a gold set; leakage controls separate the context that produced a judgment from the context in which it is evaluated; the baseline is frozen and never overwritten; a sealed development/holdout split is maintained. Quantitative eligibility requires **at least 20 adjudicated generalization-safe labels**; below that, only pilot evidence may be reported; at zero, the reported status is that accuracy improvement cannot be evaluated. Measures: assessment quality where eligible; inter-rater agreement; consistency; traceability completeness; reviewer time and cases per hour; adjudication and escalation rates; net correction; repeated-review effort; adaptation effort; the ratio of invariant to domain-specific components; and blocked-transfer reasons.

Success criteria.

- At least 20 safe adjudicated labels exist, making quantitative analysis eligible.
- The domain-specific versus transferable classification can be applied by independent raters with acceptable agreement.
- The classification has **predictive validity**: judgments it classes as transferable hold in a second context measurably better than judgments it classes as domain-specific. This is the operative test of E12.
- A common-core and adaptation map is produced under whichever plan is executed, identifying what carried over unchanged, what required adaptation, and what was blocked.
- Negative and null results are retained rather than discarded.
- Healthcare evidence appears only if every applicable gate passes.

Failure criteria — what would falsify or defeat the question.

- **No predictive validity.** If judgments classified as transferable do not hold better than those classified as domain-specific, the classification has no explanatory power. This falsifies E12's operationalization and is a substantive negative result — arguably the most informative single outcome the thesis could produce.
- **Classification not applicable.** If independent raters cannot agree on transfer status, the classification is not operational and SQ3 must be reformulated around an observable proxy rather than an origin-based criterion.
- **Transfer is uniformly negative.** If reuse degrades assessment in every second context tested, the reuse premise fails for this task — reportable, and a genuine contribution to knowing where these methods should not be applied.
- **The gate never opens.** If EXP-005 does not reach 20 adjudicated labels, SQ3 is confined to readiness evidence: the protocol, the classification scheme, and the exact statement of what is blocked. This is the stated fallback, not a silent downgrade.
- **Second setting is not genuinely different.** Under Plan B, if the "second setting" merely renames the first, the transfer test is vacuous. The contract explicitly forbids this.

4.5 The plan to answer it

Phase	Produces	Evidence yielded	Preconditions and gates	Block
S3-1 Transfer-criterion definition	A written criterion distinguishing domain-specific from general-capability uncertainty, with worked examples on both sides	The definition E12 requires; makes any later transfer claim testable	Wording closure; QL-03/QL-04 synthesis	Y1-B
S3-2 Protocol pre-registration	The pre-registered evaluation and workload protocol: hypotheses, measures, leakage controls, stopping rules, analysis plan	Protects every later result from post-hoc specification	S3-1; supervisor approval of the reviewer plan	Y1-S → Y2-A
S3-3 Reviewer recruitment and first-pass labelling	Independent reviewer 1 and reviewer 2 returns on the blind sheets	The first real independent labels; currently **0 of 2 returns**	Supervisor approval of the labelling pack; reviewers identified and available	Y1-A → Y2-A (start as early as reviewers permit)
S3-4 Adjudication	Adjudicated gold-label file; inter-rater agreement	**Opens the quantitative gate at ≥20 labels**; also unblocks SQ1 phase S1-6	S3-3 complete; adjudicator appointed	Y2-A
S3-5 Primary evaluation in software/modelling	Paired comparison; quality, consistency, traceability and effort results; validity package; traceability audit	The primary SQ3 evidence within one domain	S3-2 and S3-4; frozen baseline; no baseline overwrite	Y2-B
S3-6 Second-setting selection	The decision of which second context is used, under Plan A or Plan B, with its governance profile	Determines what kind of transfer claim is even possible	**Plan A / Plan B checkpoint** — internal control date 2026-08-26; Plan A requires all six entry gates to have a documented owner, evidence path and feasible completion date	B0 (decision) → Y2-B (setup)
S3-7 Transfer case execution	Replication in the second setting; adaptation record; blocked-transfer reasons	The transfer evidence; the test of E12's predictive validity	S3-5; second setting authorized; under Plan A, **all six entry gates passed** plus the downstream controls	Y3-A
S3-8 Common-core and adaptation analysis	Common-core/adaptation map; transfer taxonomy; domain contract; replication report; stop and fallback rules	What generalizes, what must change, when to escalate	S3-7	Y3-B
S3-9 Study 3 report	Study 3 write-up; Paper 3B (cross-domain software/modelling) or, only if every Plan A gate passed, Paper 3A (clinical informatics)	The answer to SQ3 at its supportable strength	S3-8	Y3-B → Y3-S

Phase S3-3 is deliberately placed as early as reviewer availability allows, because it gates the most downstream work of any single activity in the programme — it blocks S3-4, S3-5, S3-7, and SQ1's S1-6.

4.6 Risks and mitigations

Risk	Consequence	Mitigation
EXP-005 never reaches 20 adjudicated labels	No quantitative result for SQ3; SQ1's replay phase also blocked	Report the exact block and accept readiness-only evidence, per the contract's stated fallback. Keep the protocol pre-registered and executable. Prioritize reviewer recruitment above artifact refinement — it is the binding constraint on the whole programme
Plan A does not open	No healthcare transfer case	**Plan B is the protected default and answers every question on its own.** Study 3 replicates across another software/modelling domain, institution, dataset, diagram or model family, or reviewer panel. Plan B activates automatically if any of the six gates fails the checkpoint
Plan A opens late and destabilizes the schedule	Committed milestones displaced by a contingent extension	The reopening rule is explicit: Plan A may reopen only through completed gate evidence, downstream controls, an explicit supervisor decision, governance approval, and a schedule-impact review — and it **cannot displace committed Plan B milestones** without a recorded change
The second setting is not genuinely different	Vacuous transfer test	The contract forbids merely renaming the original setting; the second setting's difference must be stated and justified on named dimensions before S3-7
Leakage between producing and evaluating context	Inflated, invalid transfer results	Blind sheets; sealed development/holdout discipline; explicit leakage tagging; prohibition on using same-pattern judgment memory as generalization evidence
A third clinical dataset is drawn into scope informally	Gate assessment becomes incoherent	An OMOP-format dataset (EHRSHOT) was observed in a supplied Drive folder on 2026-08-11. It is **named in none of the gates, questions or plans**, all of which were written against MIMIC and Clalit. Gates 1 and 3 both require a *named* data source, so this must be resolved before either gate can be assessed. Nothing in that folder has been opened, copied or processed, and its presence changes no verdict
Reviewer effort measurement is confounded by learning effects	Effort results uninterpretable	Counterbalance case order; record session sequence; treat learning as an analysed factor rather than noise

Plan A / Plan B in full. Both plans answer the **same** main question and the same three sub-questions. Plan A extends Study 3 into a clinical guideline-adherence setting in an approved partner environment, and requires all six mandatory entry gates — use-case, people, authorization, ethics and privacy, environment, protocol — plus the downstream integrity, bounded-pilot and disclosure controls. **All six gates are currently BLOCKED; zero pass.** Plan B replicates across a second software/modelling context. The decision rule is objective: if any gate lacks a documented owner, evidence path and feasible completion date at the internal checkpoint of 2026-08-26, Plan B becomes the committed path and Plan A becomes a contingent extension. That checkpoint is an internal project-control date, not a formal university deadline.

4.7 Current status

What exists today. A deterministic policy-table comparison (`memory_informed_classifier.py`) produces a **non-destructive parallel artifact**: original classifications are preserved, only a comparison file is written, baseline behaviour change is recorded as false, and evaluation leakage is tagged. Across the four settings this produced 27 comparisons, **0** memory-informed differences, 2 human-review-after-memory flags and **0** baseline behaviour changes. A full evaluation tool-chain exists (EXP-001 through EXP-005): labelling sheets in full and blind form, an interactive labelling tool that supplies and infers nothing itself, an adjudication sheet, error analysis, accuracy summaries, paired comparison, figures, an evidence verdict and a reproducibility manifest. The expert labelling protocol specifies label values, required fields, the two-reviewer plus adjudicator workflow, and explicit bias and leakage rules. Of 27 rows, **24 are generalization-safe candidates**. A `reuse_scope` field already distinguishes four levels of reuse distance.

What is honestly missing. Zero of 24 generalization-safe expert labels are supplied; at least 20 are required. **Zero of two** reviewer returns exist. No adjudication has occurred. No transfer criterion has been written, so `E12` is currently an instruction rather than an operationalization. No second setting has been selected. The medical track is at **0 of 6** entry gates, with all three downstream controls blocked, and a third dataset has been observed that is named in no control. No artifact has been selected. The current gate status recorded by the tooling is, verbatim, that accuracy improvement cannot be evaluated yet. Consequently **no accuracy, macro-F1, generalization, effort-reduction, superiority or clinical claim is made or supportable anywhere.**

5. The three questions as one programme

The three sub-questions are not three topics. They are three stages of one lifecycle, and each hands a specific thing to the next.

	Asks	Produces	Hands to the next
SQ1	*When* should the system ask an expert?	Intervention criteria; dosage and budget policies; the coverage–load characterization	**Captured expert responses** — and, critically, a *distribution* of them shaped by the policy
SQ2	*What* must be kept, and under what governance?	Judgment representation; validation and reconciliation; provenance, scope and authority controls	**Governed, scoped judgments** with their reuse eligibility determined but not yet tested
SQ3	*Where else* does a judgment hold, and how would we know?	The domain-specific versus transferable classification; leakage-safe transfer evidence; the adaptation map	**Boundary conditions back to SQ1 and SQ2** — which uncertainties were worth asking about, and which scope rules held up

Three things about this composition are worth stating explicitly, because they are what make the programme one doctorate rather than three papers.

The handovers are typed, and the types matter. SQ1 does not hand SQ2 "some judgments"; it hands a distribution determined by the intervention policy. If the policy surfaces only hard and contested cases, the store SQ2 governs is systematically unrepresentative, and the transfer conclusions SQ3 draws inherit that skew. This is why phase U-2 specifies the interfaces in advance: the composition claim must be falsifiable, not reconstructed after the fact.

The loop closes backwards. SQ3's boundary conditions feed back into SQ1's notion of importance and SQ2's scope rules. A judgment class that never transfers is a class whose elicitation cost SQ1 should weigh differently. A scope rule that repeatedly proves too permissive is an SQ2 correction. The programme is therefore not a pipeline but a cycle traversed at least once, and the thesis reports what the return leg changed.

One shared bottleneck sits underneath all three. Independent adjudicated expert labels are required by SQ3 for transfer evidence, by SQ1 for the importance reference standard, and by SQ2 for the elicitability of scope and reasoning-responses. All three are currently gated on the same **0 of 24**. This is stated plainly rather than distributed across three risk registers, because it means one activity — reviewer recruitment and adjudication — unblocks more of the programme than any other, and should be prioritized accordingly.

Answering all three yields the main question's "captured, governed, and used." Failing any one localizes precisely which part of reliable human–AI co-reasoning remains out of reach — which is itself a reportable result, and one the proposal should name in advance rather than treat as a contingency.

6. Decisions the supervisors must make about the questions

Ten decisions already exist in the Research-Question Decision Pack; all are recorded as **Pending**. Six further decisions are **proposed by this document** and are not logged identifiers — they are numbered `D-RQ-11` onward for convenience only and should be renumbered or rejected as the supervisors see fit. One action item is included because it blocks several decisions.

ID	Decision	Bears on	Owner	Status
`A08-01`	Verify the final post-edit wording of U-RQ and SQ1–SQ3 against Ali's own saved working draft, not against the machine transcript	All four questions	Ali	**Open** — blocks `D-RQ-01`, `D-RQ-02`
`D-RQ-01`	Adopt the U-RQ wording as the working baseline, or correct it exactly	U-RQ	Iris, Arnon	**Pending**
`D-RQ-02`	Adopt SQ1 selective intervention, SQ2 governed knowledge reuse, SQ3 evaluation and transfer as the exactly-three structure	SQ1–SQ3	Iris, Arnon	**Pending**
`D-RQ-03`	Adopt the three-study mapping (one primary study per sub-question)	Programme	Iris, Arnon	**Pending**
`D-RQ-04`	Define Plan A as conditional medical transfer and Plan B as non-medical transfer	SQ3	Iris, Arnon	**Pending**
`D-RQ-05`	Confirm that all questions remain answerable under Plan B alone	All four	Iris, Arnon	**Pending**
`D-RQ-06`	Confirm 26 August 2026 as the medical-route decision gate, or replace the date	SQ3	Iris, Arnon	**Pending**
`D-RQ-07`	Accept the evidence-boundary wording as proposal language	All four	Iris, Arnon	**Pending**
`D-RQ-08`	Treat the existing literature taxonomy as the seed scope and refine the review method next	Chapter 2, all four	Iris, Arnon	**Pending**
`D-RQ-09`	Confirm the metadata-only data boundary and the continued prohibition on patient-row inspection	SQ3, Plan A	Iris, Arnon	**Pending**

`D-RQ-10`	Assign an owner to verify university candidacy rules, calendar and submission dates	Plan, all blocks	**Unassigned**	**Pending**
`D-TITLE-01`	Choose or replace the proposal title, which currently contains three words the live edit removed from the questions	Proposal	Iris, Arnon	**Pending**
`D-RQ-11` *(proposed)*	Apply one term — "expert" or "human" — consistently across U-RQ, SQ1, SQ2 and SQ3. E8 leans "expert"; U-RQ and SQ1 currently say "human"	All four	Iris, Arnon	**Proposed**
`D-RQ-12` *(proposed)*	Confirm whether "variability exploration" satisfies E6's requirement to narrow *explicitly to variability identification and classification*, or amend	U-RQ	Iris, Arnon	**Proposed**
`D-RQ-13` *(proposed)*	Retain or drop "transparently", and confirm whether `E10`'s reservation attaches to SQ2 (where the word now sits) or SQ3 (where it was recorded)	SQ2, SQ3	Iris, Arnon	**Proposed**
`D-RQ-14` *(proposed)*	Decide whether E8's capture-versus-forward-transfer split should be visible in SQ2's wording, or remain a prose distinction	SQ2	Iris, Arnon	**Proposed**
`D-RQ-15` *(proposed)*	Decide whether SQ3's wording should explicitly name the domain-specific versus broadly transferable classification, per E12	SQ3	Iris, Arnon	**Proposed**
`D-RQ-16` *(proposed)*	Decide whether U-RQ has its own artifact, or is answered by the composition of the three sub-question artifacts plus a synthesis. **Not to be decided now** — `A08-04` defers artifact work; recorded so it is not forgotten	U-RQ, Chapter 4	Iris, Arnon	**Proposed, deferred**

Recording conventions from the Decision Pack apply: an outcome is one of *Confirm*, *Confirm with correction*, *Retire or supersede*, or *Defer*. **Silence is Defer and cannot close a decision.** A deferred decision remains open and blocks readiness; it should carry an owner and a written-response deadline.

7. Standing evidence boundary

This boundary is not weakened by anything above and applies to every claim in the proposal, in any chapter, in any presentation, and in any derived document.

Independent expert labels. EXP-005 contains **0 of 24** required independent generalization-safe labels. At least **20** are required before any quantitative accuracy or generalization statement is eligible. Reviewer returns stand at **0 of 2**. Therefore **no accuracy, macro-F1, effort-reduction, generalization, or superiority claim may appear anywhere.** The tooling's own recorded status is that accuracy improvement cannot be evaluated yet. Same-pattern judgment memory may support mechanism validation only and is explicitly barred from serving as generalization evidence.

Medical track. 0 of 6 mandatory entry gates pass; all six are BLOCKED, and the three downstream controls — data integrity, bounded pilot, disclosure and export — are blocked behind them. No clinical claim, no patient-row work, no medical computation, no bounded pilot, and no export is authorized. MIMIC is an exploratory resource and is **not** the selected study dataset. A third dataset (EHRSHOT) was observed on 2026-08-11 in a supplied Drive folder; it is named in no gate, question or plan, nothing in that folder has been opened, copied or processed, and its presence changes no verdict. Permitted work is limited to governance documentation, public literature review, proposal design, stakeholder coordination, and metadata-only reconciliation that exposes no medical rows.

Literature. Searches `QL-01` through `QL-05` are protocol-ready and **frozen, not run**. No novelty claim and no review-completeness claim may be made. Every literature-gap statement in Chapter 3 and every "to our knowledge" in this document is a **candidate claim to be established or corrected** by the completed review, not a finding.

Meeting record. The 2026-08-05 record is a **machine transcript with inferred, undiarized speakers**, pending participant confirmation. Every attribution to Iris or Arnon in this document is the record's inference, not a verified quotation. No item may be marked approved solely because it is consistent with that record. The exact post-edit research-question wording is **not reliably reconstructable from the transcript** and must be taken from Ali's saved working draft (`A08-01`).

Dates. September and October 2026 targets, and the 2026-08-26 Plan A / Plan B checkpoint, are **internal working targets and project-control dates**. They are not confirmed university deadlines. The academic-calendar boundaries used for the block plan in Section 0 are unverified (`D-RQ-10`).

Plan A and Plan B. Plan A (medical extension) is conditional. **Plan B — software and modelling only — is the protected default and answers every research question on its own.** Plan A activates only through completed gate evidence, applicable downstream controls, an explicit supervisor decision, governance approval, and a schedule-impact review, and it may not displace committed Plan B milestones without a recorded change.

Question wording. All four questions are provisional pending `A08-01` verification and logged `D-RQ-01` and `D-RQ-02` decisions. Nothing in this document should be cited as an approved question.

What may be claimed today. Architecture, provenance, controlled-comparison and evaluation-readiness statements in software and modelling — nothing beyond them.